Exp2.1

LAB_EXP_DSP

Editor - C:\Users\abhiz\Desktop\LAB_EXP_DSP\EXP2.m

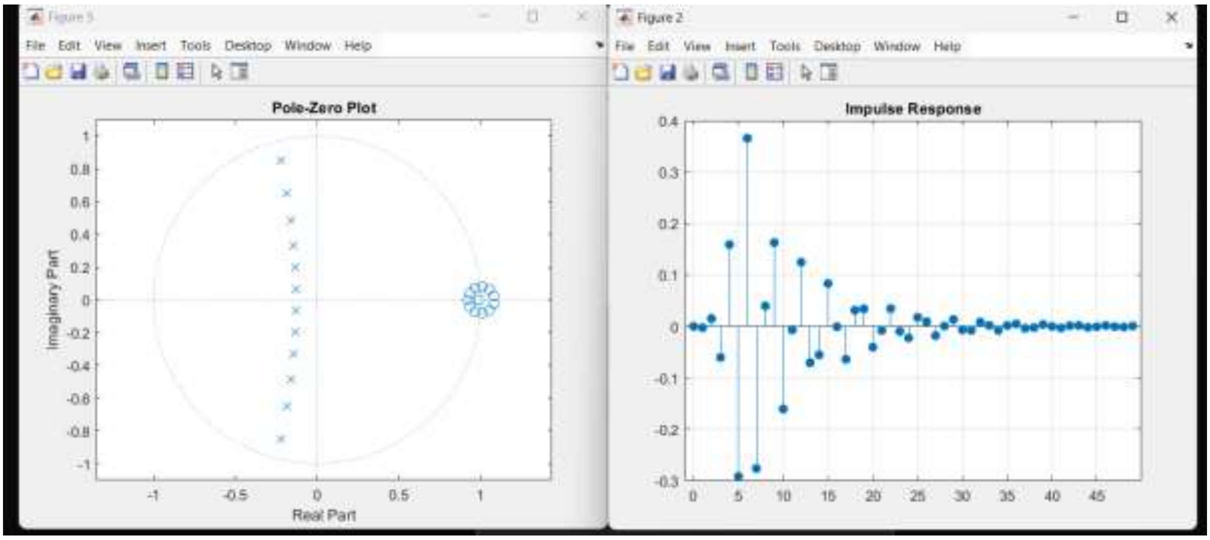
EXP2.m

```matlab
clc;
clear;
close all;

% EX NO: 1.2
% Butterworth IIR High-Pass Filter
% Design and Analysis Using MATLAB

Fs = 10000;
Fp = 3000;
Fst = 2000;
Rp = 1;
Rs = 60;

% Calculate filter order
[N,Wn] = buttord(Fp/(Fs/2), Fst/(Fs/2), Rp, Rs);

% Design Butterworth High-Pass Filter
[b,a] = butter(N, Wn, 'high');

% Magnitude and Phase Response
figure;
freqz(b,a);
```

Command Window

```
Minimum Filter Order (N) = 12
Cutoff Frequency (Wn) = 0.5807
>>
```

Output



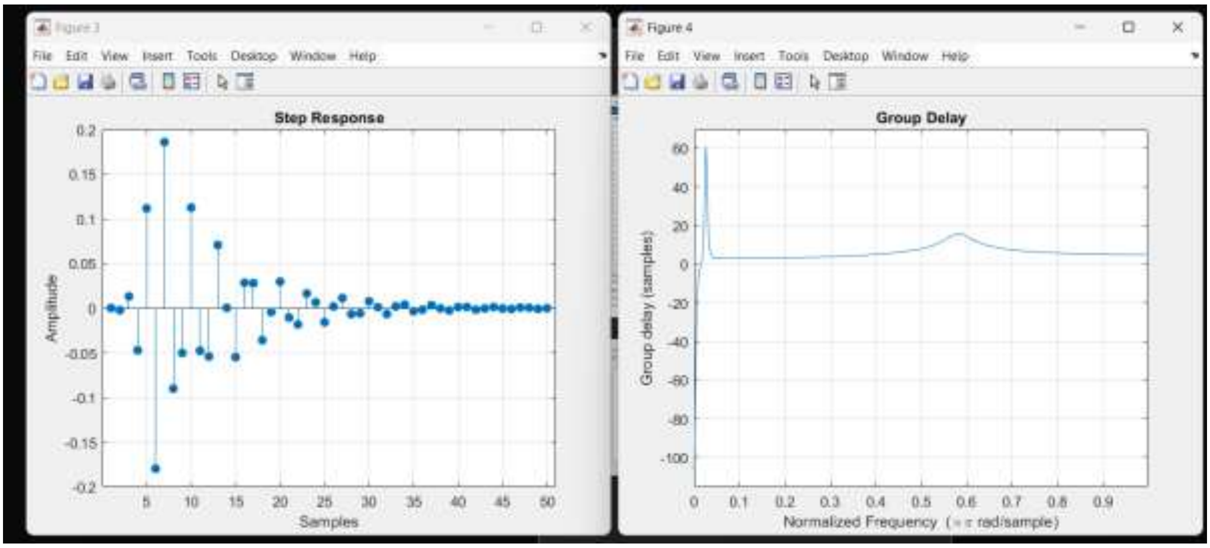


Figure 3
File Edit View Insert Tools Desktop Window Help
Step Response
Amplitude
Samples
Figure 4
File Edit View Insert Tools Desktop Window Help
Group Delay
Group delay (samples)
Normalized Frequency (×π rad/sample)

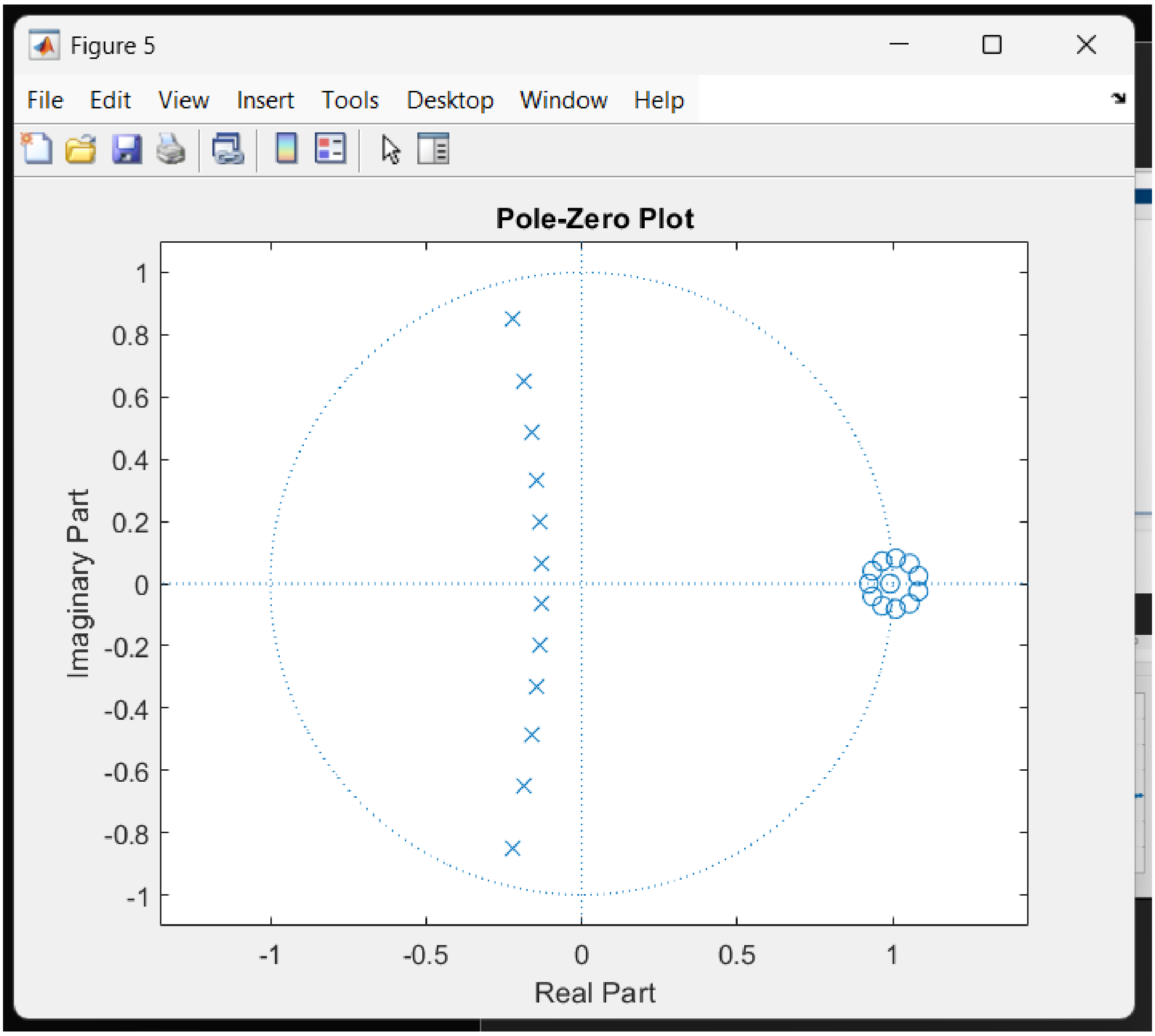
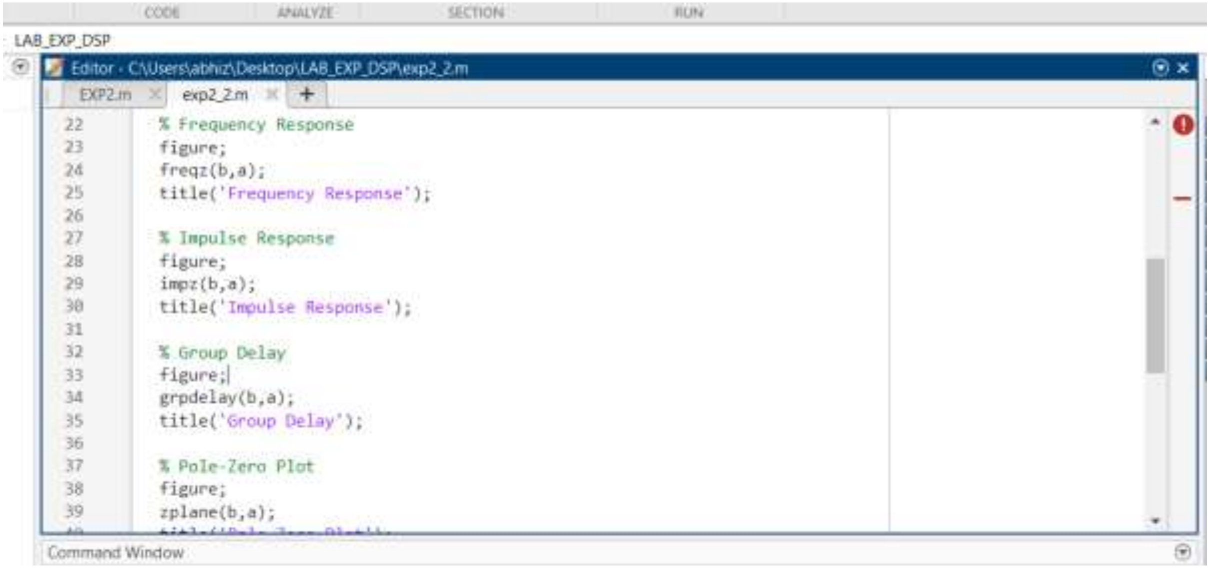


Figure 5
File Edit View Insert Tools Desktop Window Help
Pole-Zero Plot
Imaginary Part
Real Part

Exp2.2


CODE
ANALYZE
SECTION
RUN
LAB_EXP_DSP
Editor - C:\Users\abhiz\Desktop\LAB_EXP_DSP\exp2_2.m
EXP2.m
exp2_2.m
% Frequency Response
figure;
freqz(b,a);
title('Frequency Response');
% Impulse Response
figure;
impz(b,a);
title('Impulse Response');
% Group Delay
figure;
grpdelay(b,a);
title('Group Delay');
% Pole-Zero Plot
figure;
zplane(b,a);
Command Window

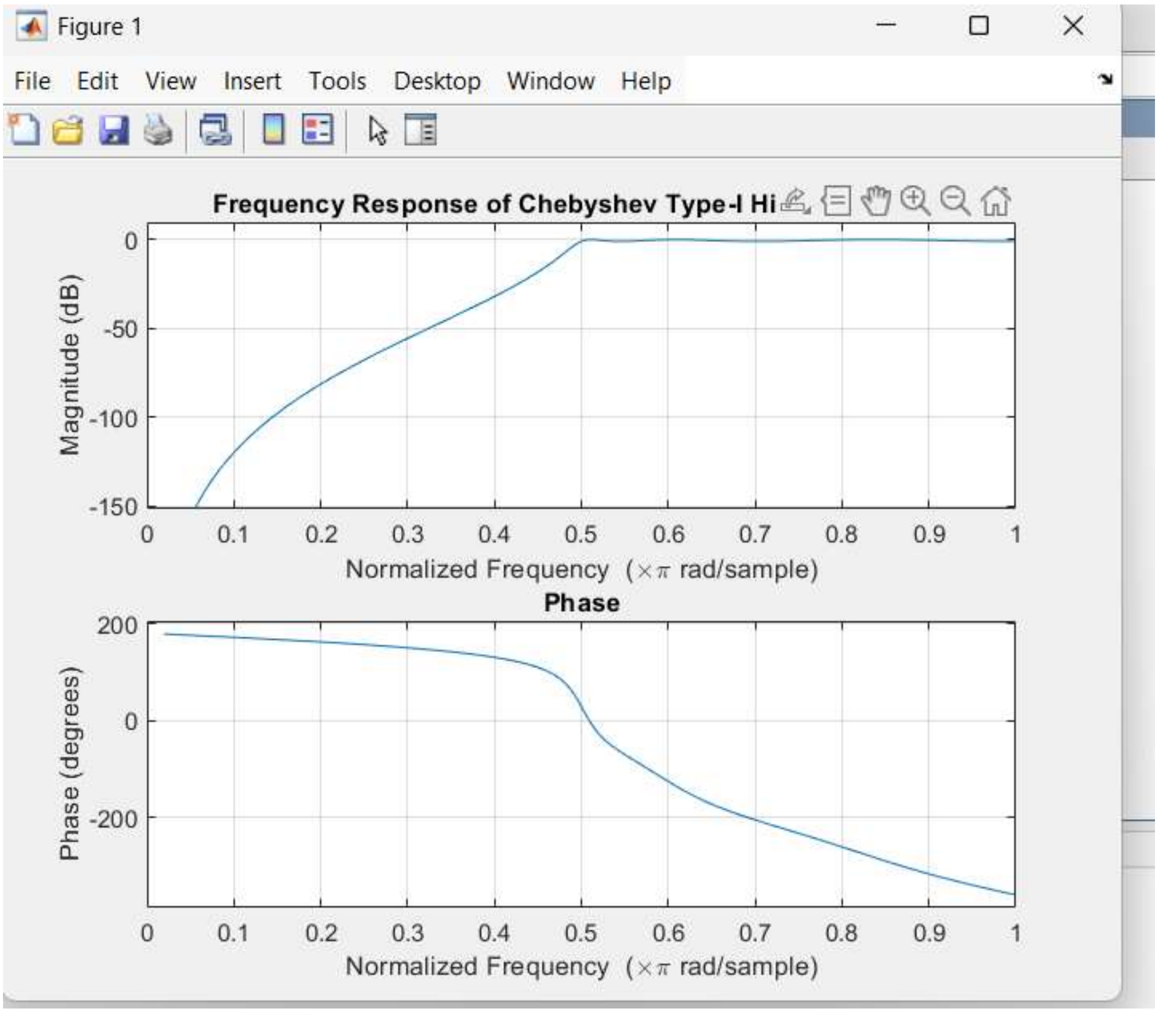
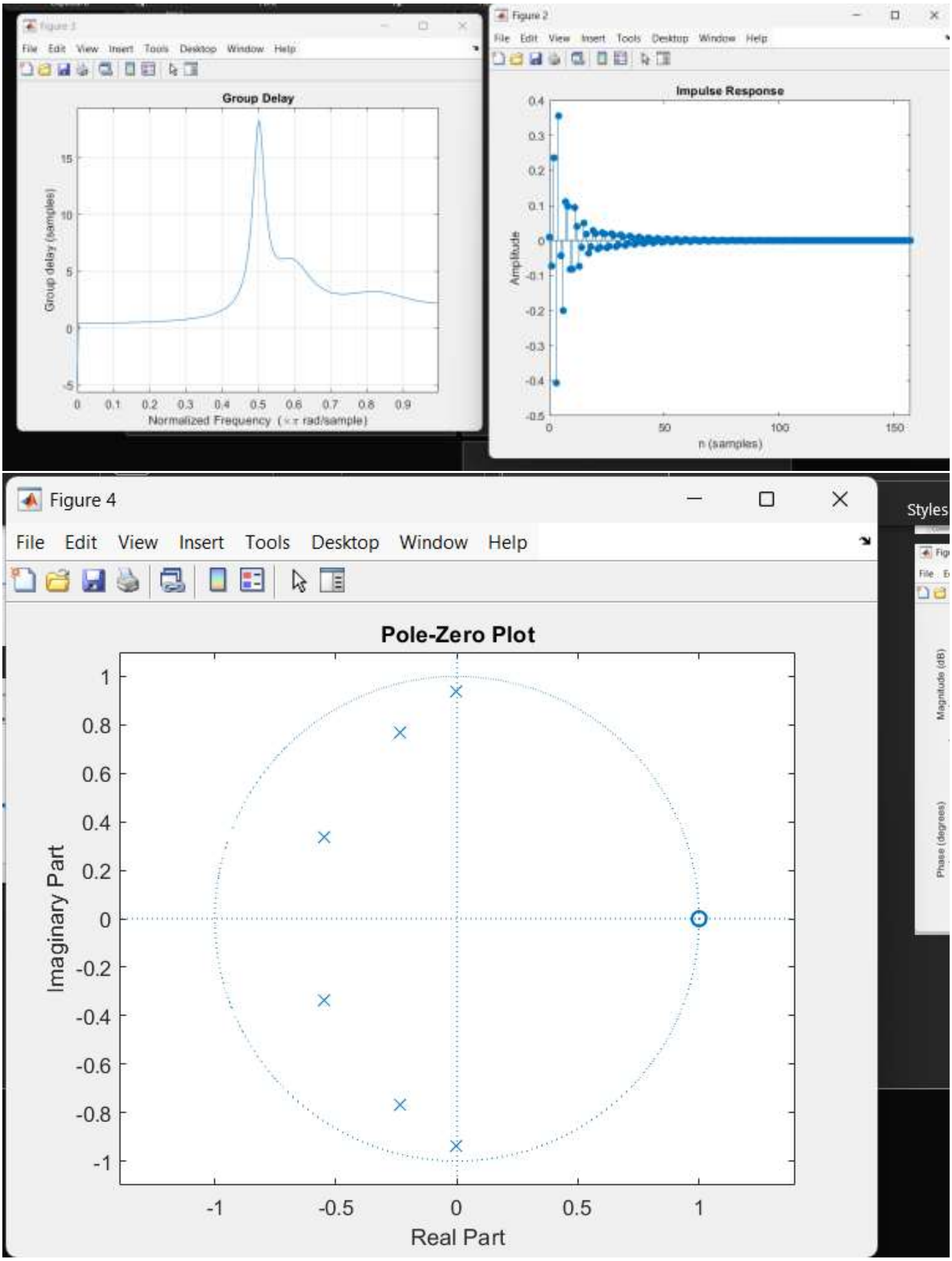


Figure 1
File
Edit
View
Insert
Tools
Desktop
Window
Help
Frequency Response of Chebyshev Type-I Hi
Magnitude (dB)
0
-50
-100
-150
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1
Normalized Frequency (×π rad/sample)
Phase
Phase (degrees)
200
0
-200
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1
Normalized Frequency (×π rad/sample)

Exp 2.3



Group Delay
Group delay (samples)
Normalized Frequency (×π rad/sample)
Figure 2
Impulse Response
Amplitude
n (samples)
Figure 4
File Edit View Insert Tools Desktop Window Help
Pole-Zero Plot
Imaginary Part
Real Part
Styles